

SPL-Monitor — Handoff & Sub-Band (63 Hz Octave) Plan

2026-06-15 · for Brian / future sessions · pick-up-cold reference

1. Where everything lives

App source	/Users/brianlloyd/Documents/Claude/Code/SPL-Monitor/ (Mac)
Live host	n8n VM 192.168.200.84 · systemd spl-monitor · /opt/spl-monitor · user brian
Public URL	https://spl.tinydoorstudios.com (Cloudflare n8n-tunnel → host :8090)
SSH	ssh -i ~/.ssh/proxmox_tds brian@192.168.200.84
SPL source	Smaart show box 192.24.143.121:26000 (up only during shows)
VM env	/etc/spl-monitor.env — SPL_SOURCE=smaart, SMAART_HOST/PORT, SPL_PORT=8090, TZ=America/New_York

Deploy (Mac → VM):

```
rsync -az --exclude .env --exclude logs --exclude __pycache__ --exclude .git \
-e "ssh -i ~/.ssh/proxmox_tds" \
/Users/brianlloyd/Documents/Claude/Code/SPL-Monitor/ brian@192.168.200.84:/opt/spl-monitor/
ssh -i ~/.ssh/proxmox_tds brian@192.168.200.84 'sudo systemctl restart spl-monitor'
```

2. What shipped this session (2026-06-15)

- **Bass Watch panel** — C-weighted track in processing.py: 6-min LCeq (from SPL C), live LCeq 10s, C–A tilt. Own lamp, WATCH mode until cYellow/cRed set. Nightly summary tracks max 6-min LCeq.
 - **Plain-language limits strip** under the hero + warn/limit labels on the chart lines.
 - **Chart y-scale fixed to 70–100 dB** (was auto from 60), clamped so >100 peaks ride the top edge.
 - **Nightly email fixed** — was failing every night since the VM move (see §6). Live send verified.
 - Doc/IP fixes: global CLAUDE.md + docs/DEPLOYED.md (VM = 192.168.200.84, not 192.168.0.125). Assets → ?v=19.
 - Reference: docs/SPL_Limits_RedRocks_vs_HollywoodBowl.pdf
- Files touched: config.json, backend/processing.py, daily.py, report.py, web/index.html, app.js, style.css, docs/DEPLOYED.md.

3. Current live state (verified)

- **Nightly email:** workflow active, cron 30 22 (22:30 EDT), HTTP → host IP :8090 → PDF+CSV → Gmail (3CDCProduction) → blloyd@3cdc.org. Verified (exec 15003 success).
- **Bass panel:** WATCH mode, currently broadband LCeq (placeholder until the octave band — this plan).
- **FSQ limits:** A warn 80 / red 90; ordinance 75 at property line; no night tiers; cYellow/cRed null.

4. Research recap (full detail in the comparison PDF)

- **C–A at FOH is a diagnostic:** ~0–3 mids/highs · ~6–12 normal music · >~15 = **bass-dominant** (watch line).
- **Off-site consensus (UK/Europe):** LCeq ≤ LAeq + 20 dB, cap **90 dB LCeq** at nearest property.
- **Paired venue limits:** Hollywood Bowl 95/108 (+13); France 102/118 (+16).
- **Red Rocks (fits FSQ):** 108 dBA 1-min; 123 dB any-band 1-min; **dB1 125 dB @ 25–80 Hz, 1-min, time-gated;** FOH-measured; 3-strikes \$10k; per-second logs tuned annually.
- **Verdict:** adopt Red Rocks' structure (sub band + night tiers + FOH→property-line + log-then-set + 3-strikes), scaled to FSQ's ~75 dBA property-line reality. Absolute amphitheater numbers don't transfer.

5. PLAN — add the 63 Hz octave sub-band metric

5.1 Decisions locked

- **Band:** 63 Hz octave (~44–88 Hz) primary (≈ Red Rocks 25–80 Hz). Optional 2nd: 31.5 Hz octave (deep EDM sub). Skip 125 Hz for limiting.

- **Window: 1-minute Leq = compliance/lamp**; 10-second = live readout. Not 15-min (sluggish), not 10s-only (too jumpy to limit on).
- **Mode:** ship in WATCH (no alarm) → log several shows across genres → set the limit from real data.

5.2 Smaart side — needs Brian

- Configure an octave-band SPL/Leq meter on the **63 Hz octave** (+ 31.5 Hz if doing two); output over the API.
- Capture the exact **API metric label(s)** Smaart streams (e.g. Leq 63 / SPL 63 Oct) — read off the All Metrics grid once live. Backend keys on the literal label.
- If Smaart can emit a 1-min band Leq directly, use it; else stream band instant/Slow and the dashboard rolls the 60s window (like the C track).

5.3 Backend (processing.py) — mirror the C track

- Add RollingLeq(10) + RollingLeq(60) fed by the band instant (60s, not 360 — LF window is 1 min).
- `_instant_sub(metrics)` picks the configured band label(s). Prefer native 1-min band Leq if present.
- `config.json` per-venue: `subBand` (label), `subWindowSec` (60), `subYellow/subRed` (null = watch).
- State fields: `subInstant`, `subShort`, `subLong`, `subYellow`, `subRed`, `subLight`, `subArmed`, `subBandLabel`, `subLongFill`. `Lamp` = `subLong` (1-min) vs `subRed`.

5.4 Frontend

- Repoint Bass panel to 1-min band Leq (big) + 10s band (live); keep C–A tilt secondary; `lamp` = `subLight`.
- Add per-neighbor sub estimate to ordinance cards (FOH band – offset). Bump asset ?v= (19 → 20).

5.5 Setting the limit (after watch period)

- Log 63 Hz 1-min Leq at FOH across genres for several shows.
- Measure a **63 Hz-specific FOH**→**property-line offset** once (pink noise / a show, handheld or 2nd input at the Westin/Davidson boundary). **Do NOT reuse the broadband offsets** — LF travels farther, so the 63 Hz drop-off is smaller.
- Set `subRed` so the worst neighbor stays under a property-line sub target. No universal single-octave number exists; anchor to UK +20/90 dBC broadband + your own complaint threshold. Derive from data.

5.6 Calibration caveat

The virtual-location offsets in `config.json` are **broadband** A/C. The 63 Hz octave offset is a different, smaller number and must be measured separately before the property-line sub estimate is trustworthy. Until then, show FOH band only and label neighbor estimates 'uncalibrated.'

6. Gotchas / operational reference

- **n8n is containerized** (bridge net `n8n_default`): workflow HTTP nodes must call `spl-monitor` at the host IP **`http://192.168.200.84:8090`**, NOT `localhost:8090`. This was the nightly-email bug. (Host→container, e.g. Slack webhook on `localhost:5678`, is fine.)
- The Gmail node statically requires the pdf attachment, so the app now always emits a PDF (minimal one-page on no-data days) via `report.build_minimal_report_pdf()`.
- **n8n 2.23.4 edit recipe:** `import:workflow` DEACTIVATES; raw DB node edits can corrupt the schedule + aren't versioned; execute `--id` can't run a Schedule-only wf (broker port 5679 clash). Reliable, as SEPARATE ssh calls: edit JSON on Mac → `scp` → `docker compose cp` → `import:workflow` → `restart n8n` → `publish:workflow --id` → `restart n8n` → `verify active=true`. Scheduler reads `workflow_entity`; cron in America/New_York. Helpers on VM: `/tmp/spl_inspect.sql`, `/tmp/wf_fixed.json`.
- VM `.env`: always write ALL vars (dropping `SPL_PORT=8090` breaks the public URL).

7. Open decisions for Brian

- One band (63) or two (63 + 31.5)?
- Bass panel: replace broadband LLeq with the octave band, or show both?
- Confirm the 1-minute compliance window.
- Provide the Smaart API metric label(s) once configured.
- (Later) the `subRed/subYellow` limit — after the watch period + a 63 Hz offset measurement.

8. Reference numbers (for limit-setting)

- C–A FOH diagnostic: ~6–12 normal · >15 bass-heavy.
- UK/Europe off-site: $L_{Ceq} \leq L_{Aeq} + 20$, cap 90 dBC at property line.
- Paired venue limits: HB 95/108 (+13) · France 102/118 (+16).
- Red Rocks: 108 dBA 1-min · 123 dB any-band 1-min · dB1 125 dB @ 25–80 Hz 1-min · FOH · 3-strikes \$10k.

Next concrete step: Brian sets the 63 Hz octave Leq in Smaart and reports the API metric label; then wire the Bass panel to a 1-minute band window (mirror the C track), ship in WATCH mode, log a few shows, measure a 63 Hz property-line offset, then set the limit.